Mental Math & Market Intuition Drills

How to Answer Every Question in 30 Seconds Even When You Don't Know

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"The candidate who says 'Let me bound this first' beats the one who goes silent."

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8.4 The Power Phrases

Memorize These Recovery Phrases

- "Let me bound this first..."
- "Rough answer: X, let me refine..."
- "I don't know the exact formula, but here's my approach..."
- "This can't be more than... and can't be less than..."
- "The intuition is... so I'd estimate..."
- "Let me sanity-check that..."

8.5 Quick Interview Cheat Sheet Module 8

Module 8: Freeze Recovery

Core Tactic: Never go silent. Bound $\rightarrow$ Estimate $\rightarrow$ Refine.
Must-Know Phrases:
- "Let me bound this first..."
- "Rough answer: X, let me refine..."
- "I don't know the exact formula, but here's my approach..."

Interview Triggers:
- Freeze? $\rightarrow$ Bound immediately
- Don't know? $\rightarrow$ Say approach, not "I don't know"
- Stuck? $\rightarrow$ Pivot to related concept

Desk Wisdom:
- Silence = death in interviews
- Process > precision
- Partial answer > no answer

Common Trap: "Don't say 'I don't know' and stop. Say 'I'd approach it by...' "
Interviewer Is Testing: Can you think under pressure?

The 3-Second Answer (Module 8)

Question: *"You freeze on a question. What do you do?"*
Your opening line:

I bound it immediately: max, min values. Then rough estimate with 0.4 rule or similar. Even if I'm 20% off, I'm showing process. Never go silentprocess beats precision in interviews.

8.6 The Non-Verbal Edge

What Interviewers Hear Beyond Your Words

Body Language Mistakes:
- Looking up ("searching memory") Signals uncertainty
- Long pauses Sounds like freezing
- Speaking faster when nervous Sounds panicked

Fixes:
- **Look at paper/screen** (not ceiling) when thinking Shows you're working, not lost
- **Narrate immediately:** "Let me bound this..." fills silence
- **Slow down 20%** when nervous Sounds more confident than you feel
- **Use hand gestures** for bounds ("Max here, min here") Visual confidence

Vocal Power Moves:

Q10: Pin Risk

Firm: Optiver (Market Maker)
Question: "You're short 1000 ATM calls, strike \$100, expiring today. Spot is \$100.02 at 3:59pm. What's your risk?"
30-Second Answer: "**Pin risk** the nightmare scenario for option sellers.
If spot closes 100.00, all calls exercise. I'm short 100,000 shares Monday morning. If spot closes < 100.00, calls expire worthless.
With spot at 100.02 and 1 minute left, it's 50/50. I have:

- 50% chance: short 100k shares (huge overnight risk)
- 50% chance: flat (no position)

Action: I need to delta-hedge NOW. Buy 50,000 shares (50% of notional) to neutralize expected assignment."
If They Probe: "Pin risk is why market makers hate ATM expiry. You can't know your position until after-hours. If I don't hedge and spot closes 100.01, I'm naked short 100k shares over weekend. If news breaks Monday, I'm destroyed. This is a career-ending mistake."

Q11-15: Rapid Fire

Q11 : "Spot \$50, vol 30%, 6M ATM call?" **A:** $0.4 \times 0.3 \times 50 \times 0.71 = 4.26$ **\$4.25**
Q12 : "Put \$5, call \$3, forward \$102, strike \$100. Arbitrage?" **A:** P - C should = K - F = -2. Actual = 2. **Yes, buy call, sell put.**
Q13 : "Why can't call value exceed spot?" **A:** Arbitrage: sell call, buy stock, pocket difference.
Q14 : "Vol smile is convex (smiles upward). Arbitrage?" **A:** No, that's normal. Concave (frown) = arbitrage.
Q15 : "Stock pays \$5 dividend tomorrow. How does \$100 strike call value change?" **A:** Drops by \$5 Œ discount factor Œ probability of being ITM. For ATM, \$2.50 drop.

11.2 FX & Cross Rates (10 Questions)

Q16: Basic Cross Rate

Firm: J.P. Morgan (FX Trading)
Question: "EURUSD = 1.10, USDJPY = 150. What's EURJPY?"
30-Second Answer: "EUR/JPY = EUR/USD Œ USD/JPY = 1.10 Œ 150 = **165.00**"
If They Probe: "This is direct multiplication because both pairs share USD. If they don't share a currency, you invert one first. Example: EURUSD and GBPUSD need to compute EURGBP = EURUSD / GBPUSD."

Q17: Triangular Arbitrage

Firm: Citadel (HFT)
Question: "EURUSD = 1.1000, GBPUSD = 1.3000, EURGBP quoted at 0.8500. Arbitrage?"
30-Second Answer: "Implied EURGBP = 1.10 / 1.30 = 0.8462 Quoted EURGBP = 0.8500 Gap = 0.8500 - 0.8462 = 0.0038 (38 pips)
Arbitrage exists.
Strategy: 1. Sell EURGBP at 0.8500 (receive 0.85 GBP per EUR) 2. Buy EUR via EURUSD at 1.10 (pay 1.10 USD) 3. Sell GBP via GBPUSD at 1.30 (receive 1.30 USD per GBP)
Profit: 0.85 Œ 1.30 - 1.10 = 1.105 - 1.10 = **0.005 USD per EUR**"
If They Probe: "In practice, bid-ask spreads kill small arb. Need 38 pips to overcome 6 pips round-trip costs. Here it's profitable. HFT algos detect this in microseconds."

Table 3: ATM Option Value Multipliers by Moneyness

Moneyness	Multiplier Range	Use This
20% OTM	0.3–0.4	0.35
15% OTM	0.4–0.5	0.45
10% OTM	0.5–0.7	0.60
5% OTM	0.7–0.9	0.80
ATM	1.0	1.00
5% ITM	1.1–1.2	1.15
10% ITM	1.3–1.5	1.40
15% ITM	1.5–1.7	1.60
20% ITM	1.7–1.9	1.80

Table 4: Discount Factors for Common Rates and Maturities

Rate	1Y	2Y	3Y	5Y
2%	0.980	0.961	0.942	0.906
3%	0.971	0.943	0.915	0.863
4%	0.962	0.925	0.889	0.822
5%	0.952	0.907	0.864	0.784
6%	0.943	0.890	0.840	0.747